

ITA06 - Machine Learning

Exp :13 Implement the Car Price Prediction Model using Python

Aim

To implement a **Car Price Prediction Model** using the **Linear Regression Algorithm** and predict the price of a car based on its features.

Software Requirements

- Python 3.x
- Jupyter Notebook / Google Colab
- NumPy
- Pandas
- Scikit-learn

Procedure

1. Import the Linear Regression model.
2. Create the dataset and target values.
3. Train the model using the training data.
4. Predict the car price for a new input.
5. Display the predicted result.

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Dataset 1: Basic Car Price Prediction

Car Age (Years) Mileage (km) Engine Size (cc) Price (₹ Lakhs)

1	12000	1200	8.5
2	18000	1200	8.0
3	25000	1300	7.4

Car Age (Years) Mileage (km) Engine Size (cc) Price (₹ Lakhs)

4	35000	1400	6.8
5	45000	1500	6.1
6	55000	1500	5.5
7	65000	1600	5.0
8	75000	1600	4.5
9	85000	1800	4.0
10	95000	1800	3.5

Target Variable

Car Price (₹ Lakhs)

New Sample

Car Age Mileage Engine Size

4 38000 1400

Dataset 2: Used Car Price Prediction

Manufacturing Year Mileage (km) Fuel Efficiency (km/L) Price (₹ Lakhs)

2024	10000	22	10.5
2023	18000	21	9.8
2022	25000	20	9.0
2021	35000	19	8.2
2020	45000	18	7.5
2019	55000	17	6.8
2018	65000	16	6.0
2017	75000	15	5.4

Manufacturing Year Mileage (km) Fuel Efficiency (km/L) Price (₹ Lakhs)

2016 85000 14 4.8

2015 95000 13 4.2

Target Variable

Car Price (₹ Lakhs)

New Sample

Manufacturing Year Mileage Fuel Efficiency

2021 32000 19

Dataset 3: Sedan Car Price Prediction

Engine Capacity (cc) Horsepower Mileage (km) Price (₹ Lakhs)

1000 70 10000 6.5

1200 82 15000 7.4

1400 95 22000 8.8

1500 105 30000 9.8

1600 115 38000 10.5

1800 130 45000 11.8

2000 145 52000 13.2

2200 160 60000 14.5

2400 175 68000 15.8

2600 190 76000 17.2

Target Variable

Car Price (₹ Lakhs)

New Sample

Engine Capacity Horsepower Mileage

1700 120 42000

Dataset 4: SUV Price Prediction

Engine Capacity (cc) Vehicle Age (Years) Distance Travelled (km) Price (₹ Lakhs)

1500	1	12000	15.5
1800	2	20000	14.8
2000	3	28000	13.9
2200	4	36000	13.0
2400	5	44000	12.0
2600	6	52000	11.0
2800	7	60000	10.2
3000	8	68000	9.4
3200	9	76000	8.7
3400	10	84000	8.0

Target Variable

Car Price (₹ Lakhs)

New Sample

Engine Capacity Vehicle Age Distance Travelled

2100 4 34000

Dataset 5: Electric Car Price Prediction

Battery Capacity (kWh) Range (km) Charging Time (Hours) Price (₹ Lakhs)

30	250	8	10.5
35	300	8	12.0
40	350	7	14.0
45	400	7	16.5
50	450	6	18.5
55	500	6	21.0
60	550	5	24.0
65	600	5	27.5
70	650	4	31.0
75	700	4	35.0

Target Variable

Car Price (₹ Lakhs)

New Sample

Battery Capacity Range Charging Time

52	470	6
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[2]

✓ Os

```
from sklearn.linear_model import LinearRegression

X = [
    [1,12000,1200],[2,18000,1200],[3,25000,1300],
    [4,35000,1400],[5,45000,1500],[6,55000,1500],
    [7,65000,1600],[8,75000,1600],[9,85000,1800],[10,95000,1800]
]

y = [8.5,8.0,7.4,6.8,6.1,5.5,5.0,4.5,4.0,3.5]

model = LinearRegression()
model.fit(X,y)

price = model.predict([[4,38000,1400]])
print("Predicted Car Price =", round(price[0],2),"Lakhs")
```

✓

Predicted Car Price = 6.8 Lakhs

[3]

✓ Os



```
from sklearn.linear_model import LinearRegression

X = [
    [2024,10000,22],[2023,18000,21],[2022,25000,20],
    [2021,35000,19],[2020,45000,18],[2019,55000,17],
    [2018,65000,16],[2017,75000,15],[2016,85000,14],[2015,95000,13]
]

y = [10.5,9.8,9.0,8.2,7.5,6.8,6.0,5.4,4.8,4.2]

model = LinearRegression()
model.fit(X,y)

price = model.predict([[2021,32000,19]])
print("Predicted Car Price =", round(price[0],2),"Lakhs")
```

✓

... Predicted Car Price = 8.06 Lakhs

```
s ▶ from sklearn.linear_model import LinearRegression

X = [
    [1000,70,10000],[1200,82,15000],[1400,95,22000],
    [1500,105,30000],[1600,115,38000],[1800,130,45000],
    [2000,145,52000],[2200,160,60000],[2400,175,68000],[2600,190,76000]
]

y = [6.5,7.4,8.8,9.8,10.5,11.8,13.2,14.5,15.8,17.2]

model = LinearRegression()
model.fit(X,y)

price = model.predict([[1700,120,42000]])
print("Predicted Car Price =", round(price[0],2),"Lakhs")

... Predicted Car Price = 11.14 Lakhs
```

```
j] Os ▶ from sklearn.linear_model import LinearRegression

X = [
    [1500,1,12000],[1800,2,20000],[2000,3,28000],
    [2200,4,36000],[2400,5,44000],[2600,6,52000],
    [2800,7,60000],[3000,8,68000],[3200,9,76000],[3400,10,84000]
]

y = [15.5,14.8,13.9,13.0,12.0,11.0,10.2,9.4,8.7,8.0]

model = LinearRegression()
model.fit(X,y)

price = model.predict([[2100,4,34000]])
print("Predicted SUV Price =", round(price[0],2),"Lakhs")

... Predicted SUV Price = 13.15 Lakhs
```

```
j] Os ▶ from sklearn.linear_model import LinearRegression

X = [
    [30,250,8],[35,300,8],[40,350,7],[45,400,7],[50,450,6],
    [55,500,6],[60,550,5],[65,600,5],[70,650,4],[75,700,4]
]

y = [10.5,12.0,14.0,16.5,18.5,21.0,24.0,27.5,31.0,35.0]

model = LinearRegression()
model.fit(X,y)

price = model.predict([[52,470,6]])
print("Predicted Electric Car Price =", round(price[0],2),"Lakhs")

... Predicted Electric Car Price = 20.72 Lakhs
```

Result

Thus, the Car Price Prediction Model was successfully implemented using the Linear Regression Algorithm. The model was trained on different car datasets and accurately predicted the prices of new vehicles based on their features such as age, mileage, engine capacity, battery capacity, fuel efficiency, and other specifications.